

TRAN NGOC MINH SONG

AI Data Engineer

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📍 HoChiMinh, VietNam

🔗 SongTran712

EDUCATION

VNU HCM University of Technology - PFIEV

- Degree: Bachelor in Electronic and Telecommunication
- GPA: 7.64/10

SKILLS

Languages: Python, C, Golang

Agent Framework: Agno, N8N, Strands Agent

Backend: Flask, FastAPI

Infrastructure: Docker, AWS

Networking: TCP/UDP, OSI model, Cellular Network Architecture, RAN and ORAN Concept

AI Coding and RnD Support: Claude Code, OpenClaw, ChatGPT

Local LLM Hosting: Ollama, vLLM, Llama.cpp

AI Finetuning/Model framework: Huggingface, Unsloth, Pytorch, Tensorflow

EXPERIENCE

AIoT Engineer

Becamex Company

📅 June 2024 – Jan 2025

📍 Binh Duong Province, Vietnam

- Study about Git, Docker, ML pipeline, EdgeX platform
- Implemented device registration and data flow management on the EdgeX IoT platform.
- Optimized edge AI inference by converting models to TensorRT, test and deploy with L4T Docker.

AI Integration Engineer

Vela Group

📅 March 2025 – March 2026

📍 Ho Chi Minh City, Vietnam

- **Agentic-RAG Chatbot:**
 - Built an AI Agent workflow in **N8N** connected with Milvus for semantic retrieval and **PostgreSQL** for structured data analysis and **GraphAPI** to connect to Facebook messenger.
 - Use **Redis** for message cache, and handle when user send text with image request
 - Demo: <https://drive.google.com/drive/u/1/folders/1C9h991wE2nnJxI7ANxXIRsymOw5-5wzE>
- **E-commerce Platform Chatbot:**
 - Crawled and extracted structured data from 1688, Alibaba, and JD using **Puppeteer** and **Crawl4AI**.
 - Developed a multilingual retrieval system with a multi-agent architecture in N8N. Core Agent for router, and sub-agents for different tasks like Product Information, Compare, Market Analysis
- **ASR Streaming/Non-Streaming with Keyword Detect:**
 - Work in 2-member build a webapp for streaming and non-streaming ASR service that this partnership need to track the keyword when their employees talk with customers
 - Main responsibility in System Architecture design, ASR task implementation
 - Use **Voice Activity Detector (VAD)** with **Zipformer lightweight model** for Streaming task
 - Use **ffmpeg**, **speaker diarization**, **Whisper-large model** for Non-streaming task
 - Handle multiple UDP audio streaming data with **Kafka**, handle load balance for ASR large model with **Redis**
 - Overview: https://github.com/SongTran712/ASR_Stream-Non-Stream_Overview-Result
- **Law Platform (still in R&D) – GraphRAG System**
 - Research and develop a **GraphRAG** system that improves traditional RAG by retrieving information based on relationships rather than only chunk similarity.

- Use **Marker** to parse PDFs into structured formats with Table of content(TOC).
- Apply LLMs to extract document structure base on TOC.
- For each header, extract relational information (e.g., Subject → Action → Object -> with Reference/Exception) and store to **Neo4J**.
- Index domain-specific data in **Milvus**.
- When receiving a query, classify it by domain, perform semantic search, and retrieve related relationships from the graph database.
- Generate final answer base on user request and context retrieve from graphDB
- Overview: <https://github.com/SongTran712/LawAgentOverview.git>
- **Excel Extraction, Standardization and Validation (still in R&D)**
 - Implement xlsx reading, mapping, extracting, header finding,... base on LLM or user defined.
 - Provide user schema, table, rule and transfer it to data schema, table schema, and SQL/formular with LLM.
 - Intergrate all service to an app help user can input flexible data and config, then receive the validation

PROJECTS

Project 1

Multimodal Classification

📅 Jan 2024 – Jun 2024

- **Tech Stack:** ReactJS, Flask, MQTT, WebSocket, Docker Compose
 - Developed a full-stack web application using ReactJS (frontend) and Flask (backend), containerized with Docker Compose for local deployment.
 - Controlled IoT devices running on **ContikiOS** via MQTT communication.
 - Designed a Transformer-based multimodal model with **Mixture of Experts (MoE)** gating to classify text, voice, and hand-pose data within a single architecture.
 - Collected custom datasets and compared multiple training strategies — fine-tuning, prompt-tuning, and feature-based learning — using pretrained models **LLaMA 3.2** and **Wav2Vec 2.0**.
 - Fine-tuned and quantized models using **Hugging Face**, **LoRA**, and **PEFT**.
 - Implemented real-time communication between frontend and backend using **WebSocket**.
 - **Paper:** <https://ieeexplore.ieee.org/document/11223472>
 - **Paper content:** <https://drive.google.com/file/d/1R3ynsrIkXOIpwo3NtmuWSw0Muzr4KK1y/view?usp=sharing>
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Project 2

Reinforcement Learning with Vision-Language Models (VLMs)

📅 Apr 2025

- Researched and explored the architecture and training principles of **Vision-Language Models (VLMs)**.
 - Trained and evaluated models on the **Cauldron** dataset for multimodal understanding.
 - Implemented a complete **RLHF (Reinforcement Learning with Human Feedback)** workflow, including base model setup, reward model design, and **PPO**-based optimization using positive/negative feedback signals.
 - **Source Code & Demo:** <https://github.com/SongTran712/ReinforceLearning-VLMs>
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Project 3

Build MultiAgent with AWS

📅 July 2025

- Lead 3-member team in **VPBank Hackathon Challenge**, implement **MultiAgent for banking process automation**.
 - Managed task allocation and integrated all AI Agent tool modules into a unified workflow (**Agent-as-Tool** and **Workflow** paradigms).
 - Use AWS Bedrock for hosting model, S3 and Opensearch for retrieving tool. Use amazon lex for intent guard.
 - Use StrandsAgent as main framework to implement multiagent workflow, deploy with Amplify, Lambda, API Gateway.
 - Certificate: https://drive.google.com/file/d/11HvSmsjHuo6A_PzHF8-oRSMZw1SAAAcA/view?usp=sharing
 - Source code: https://github.com/SongTran712/VPBank_2025
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